Systematic Search for Singularities in 3D Euler-Voigt Flows

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Agenda

Introduction

- Navier-Stokes equations ($\Omega = [0, L]^3$)

$$\begin{cases} \partial_t \mathbf{u} + (\mathbf{u} \cdot \boldsymbol{\nabla})\mathbf{u} + \boldsymbol{\nabla} p - \nu \Delta \mathbf{u} = \mathbf{0}, & \text{in } \Omega \times (0, T] \\ \boldsymbol{\nabla} \cdot \mathbf{u} = 0, & \text{in } \Omega \times (0, T] \\ \mathbf{u} = \boldsymbol{\eta} & \text{in } \Omega \text{ at } t = 0 \\ \text{Periodic Boundary Conditions} & \text{on } \partial\Omega \times (0, T] \end{cases}$$

- **The Big Question:**

  Given a smooth initial condition η, does the Navier-Stokes system always admit smooth solutions $\mathbf{u}(t)$ for arbitrarily long times t?

- One of the Clay Institute "Millennium Problems" (\$ 1M prize!)

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- $\alpha > 0$ is the regularization parameter with units of length. Acts as a 'spectral filter'. The Euler-Voigt system is globally well-posed given smooth initial conditions (Y. Cao et al. 2006).
- Conservation of the $\alpha-$Energy

$$E_\alpha(t) = \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0).$$

- As $\alpha \to 0$ the solutions of Euler-Voigt converge to the solutions of Euler and so its behaviour can be used to determine if the Euler equations will develop a finite-time singularity.

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- Theorem 5.1 (A. Larios et al. 2010).

Theorem

Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ *for some* $s \geq 3$*, let*

$$\mathbf{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$$

be the corresponding solutions to Euler and Euler-Voigt, respectively, where $0 < T < T_{max}$ *and* T_{max} *is the maximal time for which a solution to Euler exists and is unique. For all* $t \in [0,T]$*, we have*

$$\|\mathbf{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\mathbf{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \leq$$
$$\left(\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2\right) e^{Ct} + C\alpha^2(e^{Ct} - 1).$$

- Proven by Theorem 1 in (A. Larios et al. 2018).

Theorem

Let $\boldsymbol{\eta}_\alpha \in H^s$, $s \geq 3$ with $\boldsymbol{\nabla} \cdot \boldsymbol{\eta}_\alpha = 0$ and let $\boldsymbol{u}_\alpha$ be the corresponding unique solution of Euler-Voigt. Suppose that there exists a time $T^* > 0$ such that

$$\limsup_{\alpha \to 0^+} \left(\alpha \sup_{t \in [0, T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2} \right) > 0.$$

Then the Euler equations with the initial condition $\boldsymbol{\eta}_\alpha$ must develop a singularity within the interval $[0, T^*]$.

- Modified from Theorem 5.2 (A. Larios et al. 2010).

Theorem (Xinyu Zhao)

Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\Omega)$ *for some* $s \geq 3$ *satisfying*

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \textit{as } \alpha \to 0,$$

let $\mathbf{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\Omega)) \cap C^1([0,T], H^{s-1}(\Omega))$ *be the corresponding solutions to Euler and Euler-Voigt, respectively. If there exists a finite time* $T^* > 0$ *such that the solutions* $\boldsymbol{u}_\alpha$ *satisfies*

$$\limsup_{\alpha \to 0} \left(\alpha^2 \sup_{t \in [0, T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) > 0,$$

then $\mathbf{u}$ *develops a singularity in the interval* $[0, T^*]$.

- We consider analytic initial conditions that belong to the following Gevrey class

$$G^\sigma := \left\{ \mathbf{v} \in H^s(\Omega) : \|\mathbf{v}\|_{G^\sigma} = \langle \mathbf{v}, \mathbf{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0.$$

- We select the following inner product

$$\forall \mathbf{v}, \mathbf{u} \in G^\sigma, \quad \langle \mathbf{v}, \mathbf{u} \rangle_{G^\sigma} := \sum_{\mathbf{j} \in \mathbb{Z}^3} (1 + |2\pi \mathbf{j}|^2)^s e^{4\pi\sigma|\mathbf{j}|} \hat{\mathbf{v}}_{\mathbf{j}} \cdot \overline{\hat{\mathbf{u}}}_{\mathbf{j}}$$
$$= \int_\Omega (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v} \cdot \mathbf{u} \, d\mathbf{x}.$$
$$= \left\langle (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v}, \mathbf{u} \right\rangle_{L^2}$$

- The operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D|\boldsymbol{v}}\right]_{\boldsymbol{j}} := 2\pi|\boldsymbol{j}|\hat{\boldsymbol{v}}_{\boldsymbol{j}}, \qquad \left[\widehat{e^{\sigma|D|}\boldsymbol{v}}\right]_{\boldsymbol{j}} := e^{2\pi|\boldsymbol{j}|}\hat{\boldsymbol{v}}_{\boldsymbol{j}}.$$

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- We assume zero mean velocity without loss of generality.

$$V := \{\boldsymbol{v} \in G^\sigma : \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0, \quad \int_\Omega \boldsymbol{v}\, d\mathbf{x} = 0\}.$$

- We use a closed manifold $\mathcal{M}_1 \subset V$ defined as

$$\mathcal{M}_1 = \{\mathbf{v} \in V : \|\mathbf{v}\|_{\dot{H}^1} = C\}.$$

- Euler ($\mathbf{u}$) and Euler-Voigt ($\boldsymbol{u}_\alpha$) scaling property

$$\mathbf{u}_{(\alpha)}(\mathbf{x}, t) = \frac{1}{\gamma}\tilde{\mathbf{u}}_{(\alpha)}(\mathbf{y}, \tau),$$

where

$$\mathbf{y} = \beta\mathbf{x}, \tau = \lambda t, \text{ and } \frac{\beta}{\lambda\gamma} = 1.$$

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- Taylor Green Vortex ($\Omega = [0, L]^3$)

$$\boldsymbol{\eta}_{TGV} = \begin{bmatrix} V_0 \sin(2\pi x/L)\cos(2\pi y/L)\cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L)\sin(2\pi y/L)\cos(2\pi z/L) \\ 0 \end{bmatrix}$$

- Using $L = 2\pi$ and $V_0 = 1$, Bustamente et al. 2012 showed numerical evidence for finite-time singularity in Euler from $\boldsymbol{\eta}_{TGV}$ at $T^* \approx 4$. Let $\beta = \lambda = \gamma = 1$.
- We use $L = 1$ and so $\beta = 1/2\pi$. We require the same time window and so $\lambda = 1$. Therefore, $\gamma = 1/2\pi$ and so $V_0 = 1/2\pi$.

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- Calculate C analytically

$$C = \|\boldsymbol{\eta}\|_{\dot{H}^1} = \|\nabla \boldsymbol{\eta}\|_{L^2} = \left[\sum_{i,j=1}^{3} \int_{\Omega} (\partial_i \eta_j)^2 \, dx \right]^{1/2}$$

- Calculating for scaled η_{TGV}

$$\|\eta_{TGV}\|_{\dot{H}^1} = \sqrt{3}/2$$

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$$\mathcal{M}_1 = \{\mathbf{v} \in V : \|\mathbf{v}\|_{\dot{H}^1} = \sqrt{3}/2\}.$$

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- Beale-Kato-Majda (BKM) criterion (Beale et al. 1984): A Smooth solution u of the Euler system develops a singularity at $t = t_0$ if and only if

$$\lim_{t \to t_0} \int_0^t \|\omega\|_{L^\infty} d\tau = \infty, \qquad \boldsymbol{\omega} = \nabla \times \mathbf{u}$$

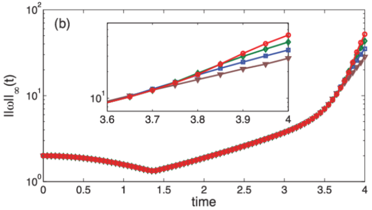

Figure: (Bustamente et al. 2012 Fig. 1b) Maximum of vorticity $\|\omega(\cdot, t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 512^3 (brown triangles), 1024^3 (blue squares), 2048^3 (green diamonds), and 4096^3 (red circles).

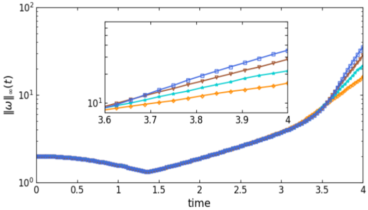

Figure: Maximum of vorticity $\|\omega(\cdot, t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), 1024^3 (blue squares).

Optimization Problem for Euler-Voigt

- Define our objective functional $\Phi_{\alpha,T} : V \to \mathbb{R}^+$

$$\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) := \|\nabla \boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{L^2}^2 = \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{\dot{H}^1}^2$$

- Given $\alpha, T \in \mathbb{R}^+$, find

$$\tilde{\boldsymbol{\eta}}_{\alpha,T} = \underset{\boldsymbol{\eta}_\alpha \in \mathcal{M}_1}{\operatorname{argmax}} \, \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha).$$

- If $\|\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T}\|_{L^2}^2 + \alpha^2 \|\nabla(\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T})\|_{L^2}^2 \to 0$ and $\alpha^2 \Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T}) > 0$ as $\alpha \to 0$ then a singularity forms in Euler the interval $[0, T]$ from initial condition $\boldsymbol{\eta}$.

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- If $\|\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T}\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T})\|_{L^2}^2 \to 0$ and $\alpha^2 \Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T}) > 0$ as $\alpha \to 0$ then a singularity forms in Euler the interval $[0, T]$ from initial condition $\boldsymbol{\eta}$.

- Project the gradient $\nabla\Phi_T(\boldsymbol{\eta}_\alpha)$ onto the tangent space to $\mathcal{M}_1$ at $\boldsymbol{\eta}_\alpha$.
- Perform vector transport operation to compute our 'momentum' term.
- Use the projected gradient and momentum term to construct a Riemannian conjugate ascent direction.
- Compute optimal step size τ_n.
- Retract back to $\mathcal{M}_1$.

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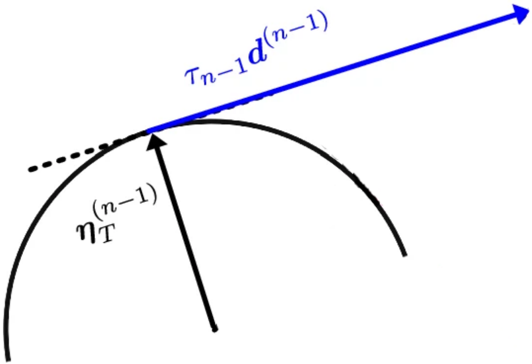
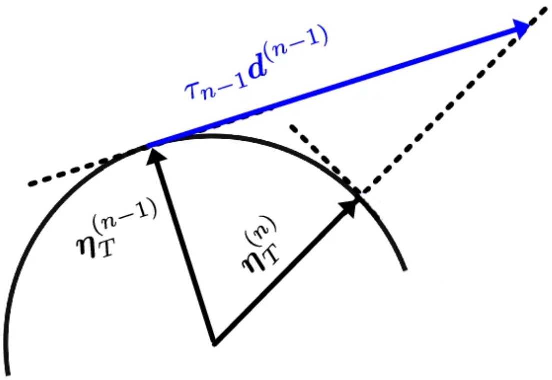


Figure: (Zhao et al. 2023, Fig. 1)



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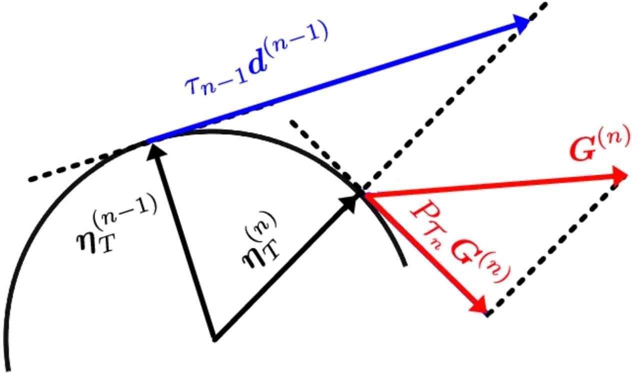

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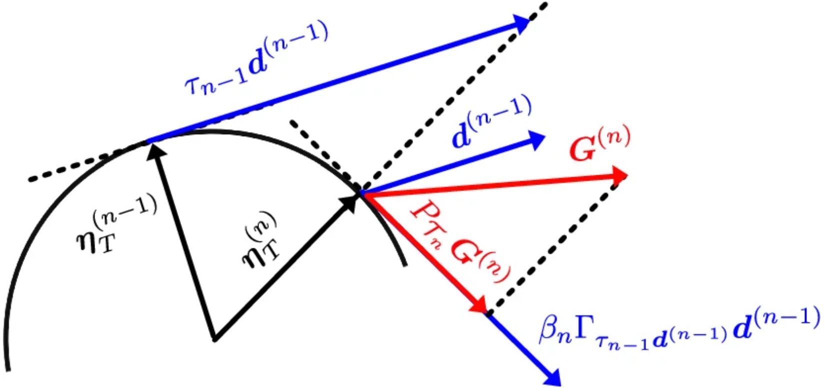

Figure: (Zhao et al. 2023, Fig. 1)

- Discretized in space using a standard pseudospectral Fourier method
- Derivatives are evaluated in the Fourier space
- Nonlinear products are computed in the physical space on an uniform grid with N points in each direction.
- Explicit fourth-order Runge-Kutta
- 2/3 dealiasing
- Domain

$$\text{resolution} : 128^3, 256^3, 512^3, 1024^3$$
$$\alpha : 1/1024, 2/1024, 4/1024, 8/1024, 16/1024$$
$$T : 3, 5$$

- Gevrey class

$$\sigma : \text{1E-5}$$
$$s : 3$$

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$$\text{resolution} : 128^3, 256^3, 512^3, 1024^3$$
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- Gevrey class

$$\begin{aligned} \sigma &: 1\text{E-}5 \\ s &: 3 \end{aligned}$$

- Relative error to $\Phi_{\alpha,T}(\eta_{TGV})$ for $dt = 2^{-8}$. Resolution: 128

	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.0045	7.00E-05
2^{-3}	Inf	Inf	0.0572	0.0011	3.64E-06
2^{-4}	0.1662	0.1157	0.0332	4.85E-04	2.02E-07
2^{-5}	0.1093	0.0715	0.0181	2.33E-04	1.21E-08
2^{-6}	0.0593	0.0369	0.0085	1.02E-04	8.63E-10
2^{-7}	0.0228	0.0137	0.003	3.45E-05	9.44E-11
2^{-8}	0	0	0	0	0

- Relative error to $\Phi_{\alpha,T}(\eta_{TGV})$ for $dt = 2^{-8}$. Resolution: 256

	1/1024	2/1024	4/1024	8/1024	16/1024
2^-2	Inf	Inf	Inf	0.003	7.00E-05
2^-3	Inf	Inf	0.0112	1.82E-04	3.64E-06
2^-4	Inf	0.0652	0.0021	9.46E-06	2.01E-07
2^-5	0.1097	0.0348	7.13E-04	5.12E-07	1.17E-08
2^-6	0.0619	0.0171	3.08E-04	3.00E-08	7.01E-10
2^-7	0.025	0.0157	1.05E-04	1.96E-09	4.04E-11
2^-8	0	0	0	0	0

- Relative error to $\Phi_{\alpha,T}(\eta_{TGV})$ for $dt = 2^{-8}$. Resolution: 512

	1/1024	2/1024	4/1024	8/1024	16/1024
2^-2	Inf	Inf	Inf	0.003	7.00E-05
2^-3	Inf	Inf	0.009	1.82E-04	3.64E-06
2^-4	Inf	0.0174	7.61E-04	9.45E-06	2.01E-07
2^-5	Inf	0.0036	3.63E-05	5.10E-07	1.17E-08
2^-6	0.0306	7.85E-04	1.72E-06	2.90E-08	7.01E-10
2^-7	0.0115	2.42E-04	8.42E-08	1.63E-09	4.04E-11
2^-8	0	0	0	0	0

- Relative error to $\Phi_{\alpha,T}(\eta_{TGV})$ for $dt = 2^{-8}$. Resolution: 1024

	1/1024	2/1024	4/1024	8/1024	16/1024
2^-2	Inf	Inf	Inf	0.003	7.00E-05
2^-3	Inf	Inf	0.009	1.82E-04	3.64E-06
2^-4	Inf	0.015	7.61E-04	9.45E-06	2.01E-07
2^-5	Inf	0.0022	3.64E-05	5.10E-07	1.17E-08
2^-6	0.0039	2.64E-04	1.71E-06	2.90E-08	7.01E-10
2^-7	5.17E-04	4.92E-06	8.36E-08	1.63E-09	4.04E-11
2^-8	0	0	0	0	0

- Maximum relative error of α-Energy in $[0, T]$. Resolution: 128

	1/1024	2/1024	4/1024	8/1024	16/1024
$2^{-}2$	Inf	Inf	Inf	3.82E-04	8.73E-06
$2^{-}3$	Inf	Inf	1.94E-03	6.91E-05	3.72E-07
$2^{-}4$	3.38E-03	2.74E-03	1.11E-03	3.05E-05	1.76E-08
$2^{-}5$	2.38E-03	1.81E-03	6.49E-04	1.57E-05	9.59E-10
$2^{-}6$	1.51E-03	1.09E-03	3.56E-04	8.05E-06	7.55E-11
$2^{-}7$	8.67E-04	6.06E-04	1.87E-04	4.08E-06	1.47E-11
$2^{-}8$	4.69E-04	3.21E-04	9.59E-05	2.05E-06	6.00E-12

- Maximum relative error of α-Energy in $[0, T]$. Resolution: 256

	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	4.34E-04	1.49E-05	3.72E-07
2^{-4}	Inf	8.36E-04	5.61E-05	6.85E-07	1.75E-08
2^{-5}	9.63E-04	4.35E-04	1.80E-05	3.40E-08	9.09E-10
2^{-6}	6.33E-04	2.48E-04	8.97E-06	1.89E-09	4.95E-11
2^{-7}	3.83E-04	1.37E-04	4.59E-06	1.38E-10	2.89E-12
2^{-8}	2.14E-04	7.23E-05	2.33E-06	2.37E-11	2.95E-12

- Maximum relative error of α-Energy in $[0, T]$. Resolution: 512

	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	3.85E-04	1.49E-05	3.72E-07
2^{-4}	Inf	2.72E-04	2.58E-05	6.85E-07	1.75E-08
2^{-5}	Inf	2.89E-05	1.12E-06	3.38E-08	9.09E-10
2^{-6}	1.58E-04	8.18E-06	5.00E-08	1.82E-09	4.95E-11
2^{-7}	9.02E-05	3.74E-06	2.47E-09	1.03E-10	3.44E-12
2^{-8}	4.98E-05	1.91E-06	1.42E-10	6.31E-12	2.69E-12

- Maximum relative error of α-Energy in $[0, T]$. Resolution: 1024

	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	3.85E-04	1.49E-05	3.72E-07
2^{-4}	Inf	2.53E-04	2.58E-05	6.85E-07	1.75E-08
2^{-5}	Inf	2.67E-05	1.12E-06	3.38E-08	9.09E-10
2^{-6}	1.46E-05	1.26E-06	4.99E-08	1.82E-09	4.95E-11
2^{-7}	2.53E-06	5.12E-08	2.45E-09	1.03E-10	3.25E-12
2^{-8}	1.06E-06	2.35E-09	1.30E-10	6.31E-12	2.78E-12

Results

- Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$ for each resolution at $T = 3, 5$.

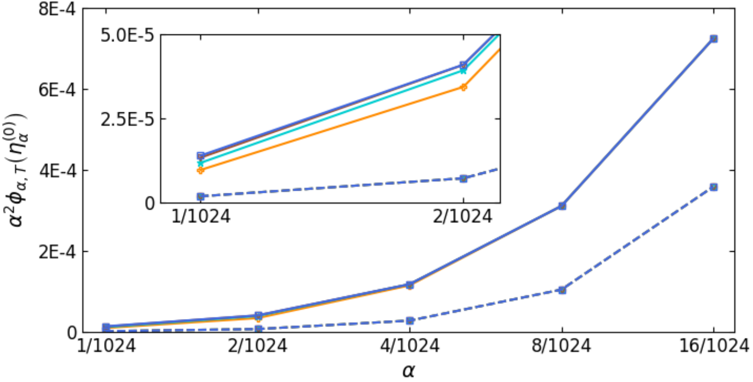

Figure: Results for $T = 3$ (dotted line) and for $T = 5$ (solid line). Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), and 1024^3 (blue squares).

- Ansatz 4.2 (Larios et al. 2018)

$$\sup_{t\in[0,T^*]} \|\nabla \boldsymbol{u}_\alpha\|_{L^2} \approx \mathcal{O}(\alpha^p)$$

.

- Singularity formation criteria

$$\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}^0_\alpha) > 0 \text{ as } \alpha \to 0$$

- Singularity formation occurs if $p \leq -1$.

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- Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. T=3.

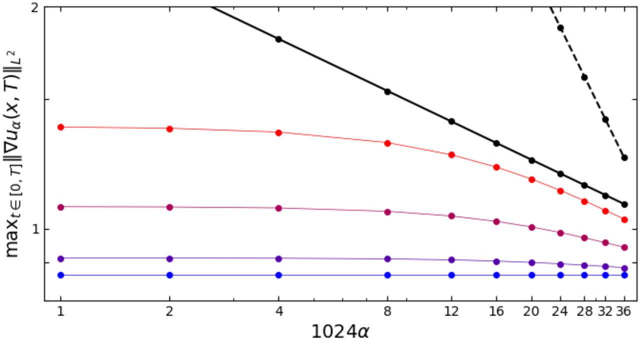


Figure: Log-log plot of $max_{t\in[0,T*]}\|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs α at $T=0$ (blue) to $T=3.0$ (red). $C\alpha^p$ (black), $p=-0.234$. $C\alpha^{-1}$ (dotted black).

- Results of $\alpha^2 \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. T=5.

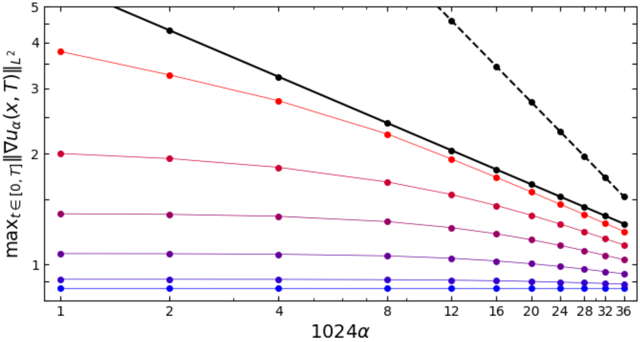


Figure: Log-log plot of $max_{t\in[0,T*]} \|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs α at $T=0$ (blue) to $T=5.0$ (red). $C\alpha^p$ (black), $p=-0.417$. $C\alpha^{-1}$ (dotted black).

- Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$.

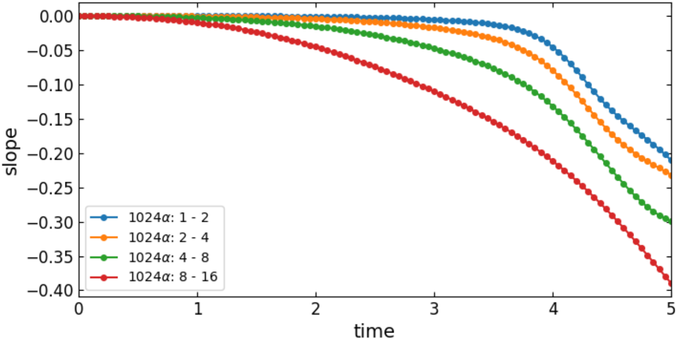
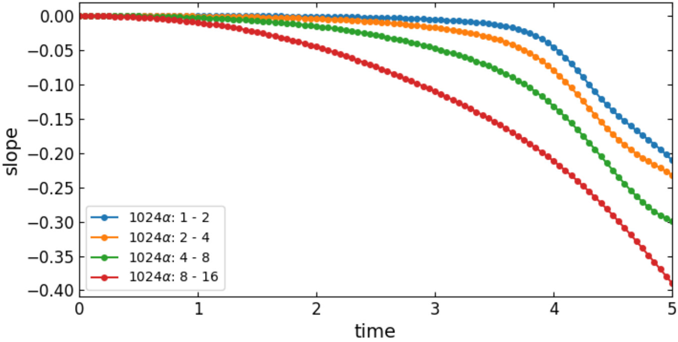


Figure: Plot of the slope of $max_{t\in[0,T^*]}\|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs *time*.

- Therefore $p = -0.138$ for $T = 3$ and $p = -0.390$ for $T = 5$.
- Our pre-optimization results do not indicate that singularity formation will occur from the Taylor Green Vortex.

- Results of $\alpha^2 \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$.



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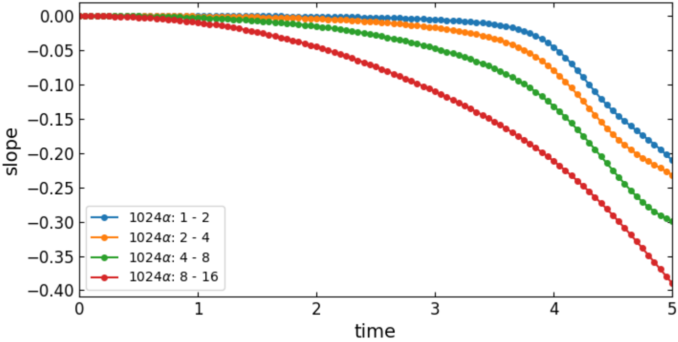

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- Maximization of $\Phi_{\alpha,T}(\boldsymbol{\eta}_{\alpha}^{(n)})$ vs n. Resolution: 128, T=3.

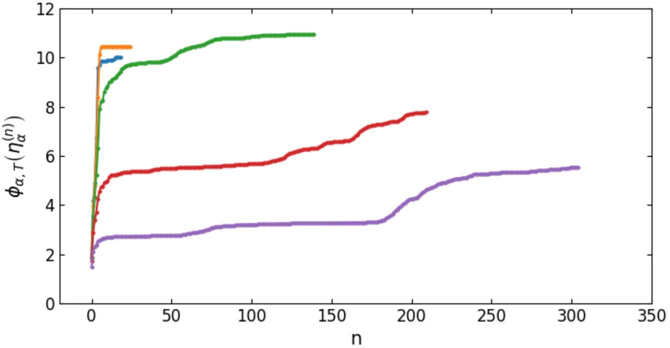


Figure: α : 16/1024 (purple), α : 8/1024 (red), α : 4/1024 (green), α : 2/1024 (orange), α : 1/1024 (blue).

- Maximization of $\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^{(n)})$ vs n. Resolution=128, T=5.

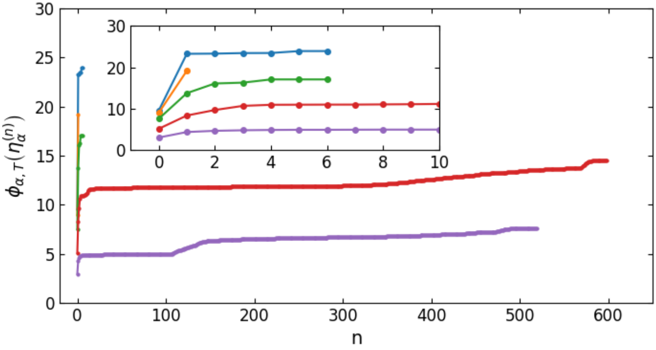


Figure: α : 16/1024 (purple), α : 8/1024 (red), α : 4/1024 (green), α : 2/1024 (orange), α : 1/1024 (blue).

- Results of $\alpha^2 \Phi_{\alpha,T}(\tilde{\eta}_{\alpha,T})$ as $\alpha \to 0$. Resolution=128.

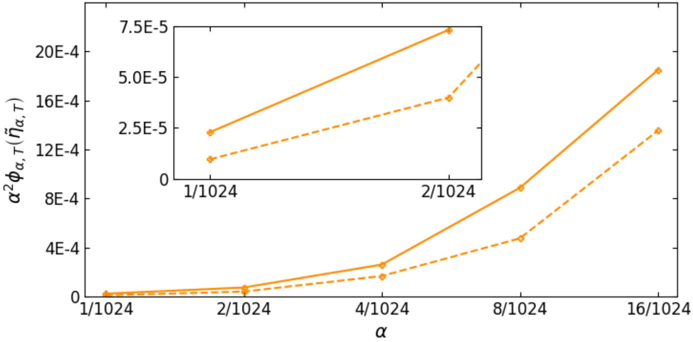


Figure: Results for $T = 3$ (dotted line) and for $T = 5$ (solid line). Results from runs performed at 128^3 (orange plus).

- Results of $\alpha^2\Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T})$ as $\alpha \to 0$. Resolution=128, T=3.

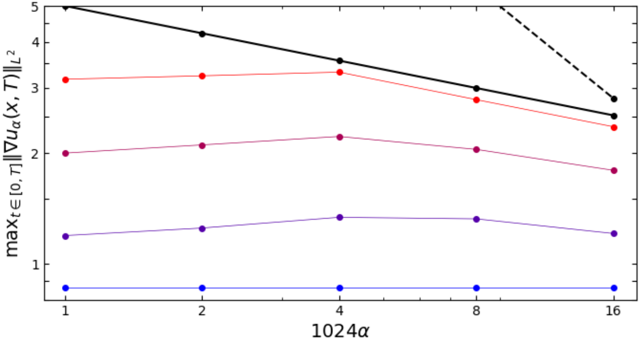


Figure: Log-log plot of $max_{t\in[0,T*]}\|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs α at $T=0$ (blue) to $T=5.0$ (red). $C\alpha^p$ (black), $p=-0.246$.

- Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. Resolution=128, T=3.

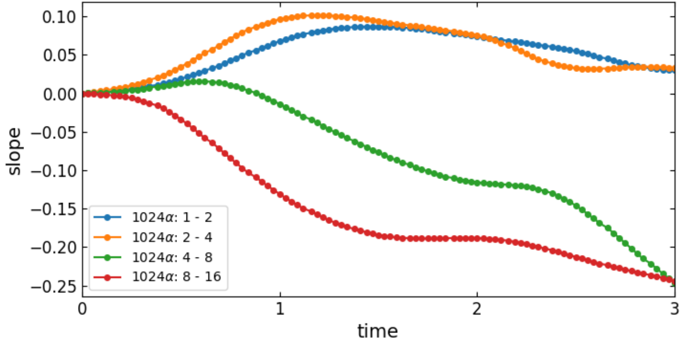
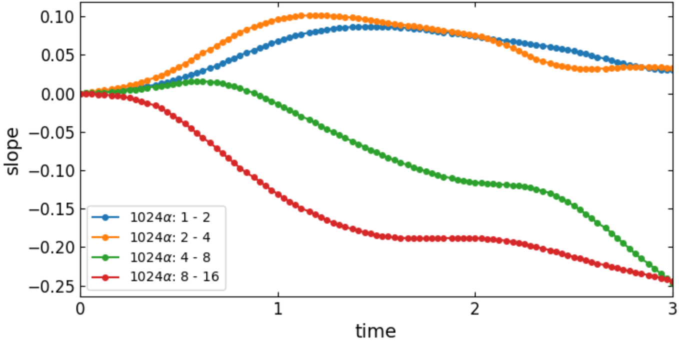


Figure: Plot of the slope of $max_{t\in[0,T*]}||\nabla \boldsymbol{u}_\alpha||_{L^2}$ vs *time*.

- Therefore $p = -0.246$ and so singularity formation does not occur.

- Results of $\alpha^2 \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. Resolution=128, T=3.



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- Therefore $p = -0.246$ and so singularity formation does not occur.

- Results of $\alpha^2\Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T})$ as $\alpha \to 0$. Resolution=128, T=5.

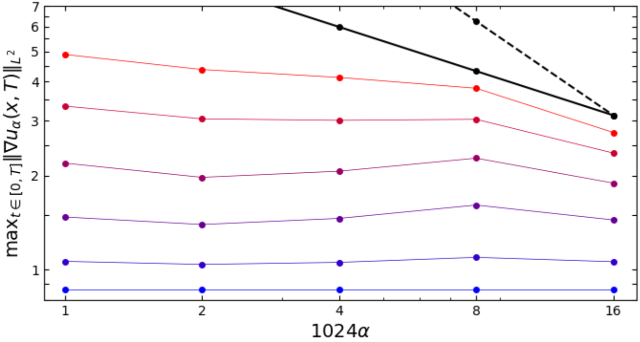

Figure: Log-log plot of $max_{t\in[0,T*]}\|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs α at $T=0$ (blue) to $T=5.0$ (red). $C\alpha^p$ (black), $p=-0.470$.

- Results of $\alpha^2 \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. Resolution=128, T=5.

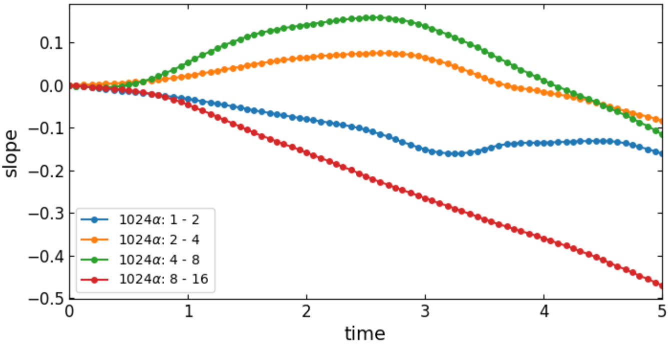

Figure: Plot of the slope of $max_{t\in[0,T^*]} \|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs *time*.

- Therefore $p = -0.470$.
- Our optimization results do not indicate that singularity formation will occur for a resolution of 128.

- Results of $\alpha^2 \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. Resolution=128, T=5.

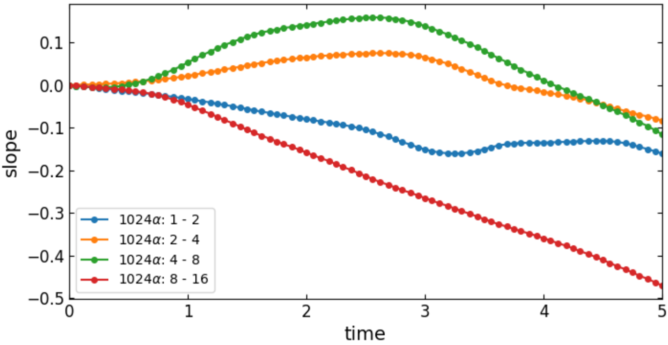

Figure: Plot of the slope of $max_{t\in[0,T^*]}\|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs *time*.

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- Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. Resolution=128, T=5.

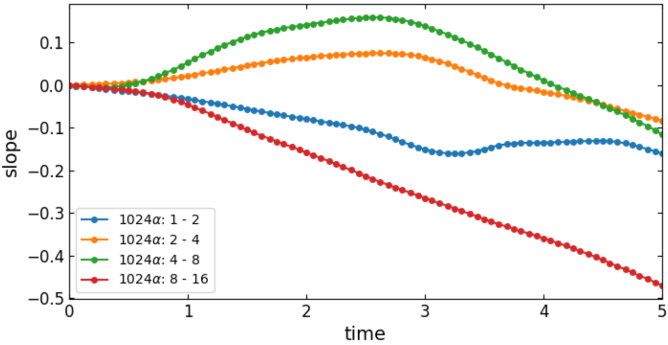

Figure: Plot of the slope of $max_{t\in[0,T^*]}\|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs *time*.

- Therefore $p = -0.470$.
- Our optimization results do not indicate that singularity formation will occur for a resolution of 128.

- Results of $\alpha^2\Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T})$ as $\alpha \to 0$. Resolution=256.

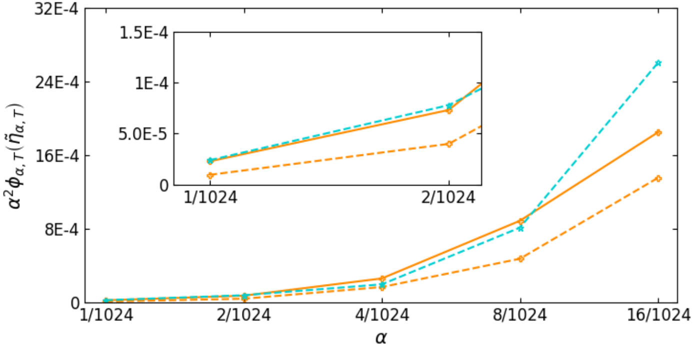

Figure: Results for $T = 3$ (dotted line) and for $T = 5$ (solid line). Results from runs performed at 128^3 (orange plus), 256^3 (cyan star).

- Results of $\alpha^2\Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T})$ as $\alpha \to 0$. Resolution=256, T=3.

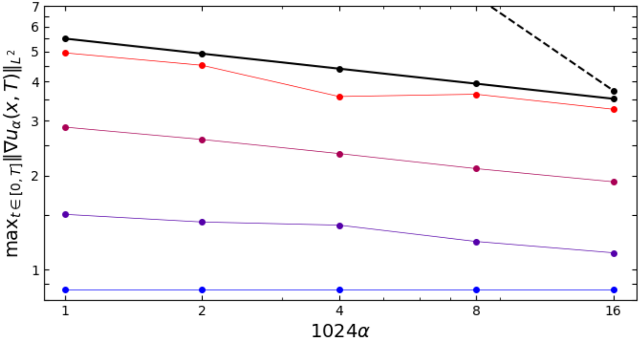

Figure: Log-log plot of $max_{t\in[0,T*]}\|\nabla \boldsymbol{u}_\alpha\|_{L^2}$ vs α at $T=0$ (blue) to $T=5.0$ (red). $C\alpha^p$ (black), $p=-0.160$.

- Our results indicate that without optimization, singularity formation will not occur in the Euler equations with the Taylor Green Vortex initial condition.
- Upon low resolution optimization, the results improve, but still do not indicate singularity formation.
- Our next step is to repeat with higher resolutions.
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Thank You